

Jadd-Ilyes Ali Larbi

Mechanical Engineer

DETAILS

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Linked in

EDUCATION

EPFL, MSc in Mechanical Engineering
2023 – 2026

EPFL, BSc in Mechanical Engineering
2020 – 2023

SKILLS

CAD/CAE: 3DEXPERIENCE, CATIA, SolidWorks

Structural Analysis: Nonlinear FEA, composite structural modeling, laminate sizing, assembly-level simulation, failure criteria (Hashin, Tsai-Wu), result interpretation and validation

Simulation & Testing: Physics-based modeling, simulation-to-test assessment, experimental validation, DAQ, CFD (ANSYS Fluent)

Manufacturing: Composite design & layup, prototyping, CNC milling

Programming: MATLAB, Python, C

LANGUAGES

French : native
English : C1
German : B1 (learning)
Spanish : B1 (learning)
Chinese : A1

INTERESTS

Strength training & endurance sports
FPV drone flying
Photography
Language learning & international travel

Profile

Mechanical engineer (EPFL) specializing in structural and composite engineering, with hands-on experience in nonlinear FEA, system integration, and full-scale hardware development. Delivered end-to-end engineering solutions on electric, hybrid, and hydrogen mobility projects at Navier and Swiss Solar Boat, from physics-based modeling and design to manufacturing, testing, and validation. Strong CAD/CAE background (3DEXPERIENCE, CATIA, SolidWorks) and a rigorous approach to model assumptions, result interpretation, and simulation limits.

Experience

Navier Inc. – Mechanical Engineering Intern

6 months – Oct 2025 – April 2026 – Alameda, San Francisco Bay Area, CA, USA

- Designed and integrated key subsystems of a 160 hp ICE-powered range-extender generator for a hybrid electric boat, including mechanical/composite design, packaging, and environmental sealing of the electric motor
- Performed nonlinear structural analyses on full-scale hydrofoils (1.5 m foils, 2 m struts) in 3DEXPERIENCE to establish realistic safety factors and stiffness/weight trade-offs, supporting redesign decisions
- Led a simulation-to-test correlation study on hydrofoil structures; quantified combined uncertainties across geometry, laminates, boundary conditions, loads, and instrumentation, and demonstrated that several configurations could not be validated as designed – driving redesign decisions on critical components
- Developed a physics-based deflection prediction model for composite parts (implemented in Excel), improving accuracy by 84% over the existing in-house method; used for sizing and validation of manufactured parts.
- Designed, sourced materials for, manufactured, and validated walkable composite access panels (~2 m) above four 28.5 kWh battery packs for technician access and safety
- Owned simulation and composite topics across CAD, prototyping, manufacturing, validation, and supplier coordination in a ~30-person multidisciplinary team

Swiss Solar Boat – EPFL Engineering Project

3 years – 2023 – 2025 – Lausanne, Switzerland

- Led a 12-person mechanical and hydrodynamics team developing foils, propulsion systems, and structural solutions for a hydrogen-powered vessel
- Redesigned the hull in 3DEXPERIENCE to integrate a 200 kg fuel cell and 100 kg H_2 tank while maintaining mechanical integration and static/dynamic stability
- Built a full-vessel composite structural model covering hull, foils, and appendages,120 composite surfaces, 50 structural volumes, 10 load cases, for laminate sizing and structural validation
- Defined laminate architectures and released 70+ composite parts manufactured with NTPT (the prepreg supplier used in F1 and high-end watchmaking), in collaboration with Decision SA (Alinghi America's Cup boatbuilder)
- Recognized as the FEA / composite simulation reference within the team and consulted by other EPFL engineering associations for methodology, modeling, and laminate analysis support
- Manufactured a 7-meter carbon-fiber boat end-to-end and engineered torpedo propulsion units, including epicyclic gear train sizing (KISSsoft), gearbox housing, packaging, and reliability work

Academic Distinctions & Additional Experience

- Valedictorian in Mechanical Physics (ranked 1st out of 400 students)
- Teaching Assistant (4 years) in Mechanical Physics, Thermodynamics, and Electromagnetism